CSE 4106: Computer Networks Laboratory

OSPF-TE VS RIP ROUTING PROTOCOLS

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1 Objective

The main goal of this project is to use OMNeT++ to design and carry out a full network simulation that compares how well two dynamic routing protocols work: OSPF with Traffic Engineering (OSPF-TE) and Routing Information Protocol (RIP). We want to use this simulation to look at how these protocols work and what they are like in different network situations. We will pay special attention to how well they work with different types of traffic, such as video streaming, voice over IP, and data transfer apps.

The project aims to furnish empirical evidence concerning the influence of Traffic Engineering on network optimization and the delivery of Quality of Service (QoS). We can set clear performance benchmarks by systematically comparing the two protocols' convergence times, throughput capabilities, latency characteristics, and packet loss rates. The study also looks at how different protocol parameters affect the performance of the whole network, giving us information that can help us make decisions about how to design networks in the real world. This comparative analysis is particularly pertinent for discerning when contemporary traffic-engineered protocols confer substantial advantages over conventional distance-vector routing methodologies.

2 Project Description

This network simulation project uses a hybrid network topology with two separate routing domains: an OSPF-TE network and a RIP network. OMNeT++ version 6.2.0, a discrete event simulation framework that is commonly used in network research, was used to build the simulation environment. The design includes many network parts that have been carefully made to accurately model how modern networks work and how traffic flows.

2.1 Network Architecture

The simulated network has four end host nodes, called host1 through host4, that act as both traffic sources and destinations during the simulation. These hosts are connected to two separate routing networks that work independently. This lets you directly compare the performance of the routing protocols.

The OSPF-TE network is the main part of the simulation. It has five routers that are connected in a mesh topology. These routers use high-capacity links that work at 1 Gigabit per second and have propagation delays of 2 to 3 milliseconds. This setup is an example of modern high-performance networking infrastructure and serves as the basis for showing off advanced traffic engineering skills. The mesh topology lets the OSPF-TE protocol use its advanced path selection algorithms to make the best routing decisions by giving it many ways to get from one point in the network to another.

The RIP network, on the other hand, has four routers connected by slower links that work at 100 megabits per second and have a propagation delay of 5 milliseconds. This setup is like older network infrastructure or smaller deployments where simpler routing protocols are still used. The intentional differences in capacity and latency between the two networks make it possible to make meaningful performance comparisons in real-world situations.

The simulation has three special traffic generators that create patterns for video, voice, and data traffic. Each generator has been set up with settings that are similar to those of real-world applications. This makes sure that the simulation results are accurate representations of how networks work in the real world. A performance monitoring module also collects and analyzes network metrics throughout the simulation, giving us a lot of information to compare protocols.

2.2 Traffic Engineering Skills

The OSPF-TE implementation has a very advanced Traffic Engineering Database (TED) that keeps track of the current state of network links, such as the amount of bandwidth available, the propagation delay, and the current load levels. This database is updated all the time at set intervals, so routers can make routing decisions based on the current state of the network instead of just static metrics.

Constraint-Based Routing (CBR) is an important part of how OSPF-TE works. CBR looks at paths based on more than just hop count or basic link costs. It looks at multiple factors at once. The system can make sure that the chosen paths meet the Quality of Service needs of different types of traffic by enforcing minimum bandwidth and maximum delay limits. This feature is especially useful for multimedia apps that need to work well under strict conditions.

The simulation can handle many different operational scenarios, such as high load conditions, low latency optimization, high throughput maximization, and load balancing across multiple paths. Each scenario has been carefully set up to test different parts of the routing protocols, showing their strengths and weaknesses. The simulation's ability to be changed makes it possible to do a lot of parameter studies, which helps researchers figure out how different protocol settings affect the performance of the whole network.

3 Project Files

This part gives a full list of all the project files, sorted by what they do in the simulation framework. To fully understand the system architecture and implementation approach, you need to know what these parts do and how they work together.

3.1 Network Description Files

The `package.ned` file is the first file that defines the network topology. It is also the namespace identifier for all of the project's modules. This structure follows OMNeT++ rules and makes sure that modules are resolved correctly when they are compiled and run.

The `NetworkTopology.ned` file shows how all the parts of the network are connected to each other and how they work together. This file makes four end hosts, five OSPF-TE routers, four RIP routers, three traffic generators, and one performance monitor. The OSPF-TE routers are set up in a pentagon mesh shape, and there are extra cross-connections between them to give you more options for path redundancy and load balancing. This network has links that work at 1 Gigabit per second and have propagation delays of 2 to 3 milliseconds. This is an example of high-performance infrastructure for businesses or service providers.

The RIP network topology uses a square shape, and all of the routers are fully connected to each other. These links run at 100 megabits per second with a 5-millisecond propagation delay, which is like how small to medium-sized business networks or old infrastructure would work. The connections between hosts and their routers work at 100 Megabits per second with very little delay (1 millisecond), which is normal for access layer connectivity.

3.2 Router Module Definitions

The `OSPFTERouter.ned` file tells OSPF-TE routers what their interface and settings should be. Each router has its own unique ID and works in a specific OSPF area. The default backbone area is area 0.0.0.0. The Traffic Engineering metric is made up of three weighted parts: bandwidth usage, propagation delay, and current link load. You can change these weights to make routing work better for different types of traffic or operational needs.

Constraint-based routing works with minimum bandwidth and maximum delay thresholds that can be changed. By default, these are set to 50 Megabits per second and 100 milliseconds, respectively. These limits make sure that the paths that are chosen can handle applications that need high quality. There are hello intervals of 10 seconds for finding neighbors, dead intervals of 40 seconds for finding out when a neighbor has timed out, LSA intervals of 30 seconds for sending out link state advertisements, and TED update intervals of 5 seconds for refreshing traffic engineering information.

The `RIPRouter.ned` file explains how to set up the simpler distance-vector routing protocol. RIP routers keep routing tables based on hop count metrics and share routing information at set times. The default update interval of 30 seconds is based on the RIP Version 2 specification, and route timeout mechanisms make sure that old routing information is eventually removed from routing tables. The limit of 15 hops on the maximum hop count stops problems with counting to infinity that are built into distance-vector protocols.

3.3 Host and Traffic Generator Modules

The `SimpleHost.ned` file sets up basic end-system functions, such as where traffic generators can connect and where generated traffic will end up. These hosts can both send and receive network traffic because they have simple receive-and-acknowledge behavior.

Video traffic generation follows the same patterns as most modern streaming apps. The `VideoTrafficGenerator.ned` module sends out big packets of 1500 bytes every 30 frames per second, which is what standard video streaming needs. The configured bitrate of 5 Megabits per second is good for medium-quality video streaming. This is good for things like video conferencing and entertainment streaming services. This traffic pattern puts a lot of strain on the network's bandwidth and tests the routing protocols' ability to find paths with a lot of capacity.

The `VoiceTrafficGenerator.ned` module makes voice traffic that is like Voice over IP (VoIP) applications. VoIP sends small packets of 160 bytes at a high rate, 50 packets per second, which is different from video traffic. The set bitrate of 64 kilobits per second is in line with the standard G.711 codec specifications. VoIP traffic is very sensitive to delay and jitter, so it's a great way to test the QoS capabilities of routing protocols. The 5-millisecond jitter tolerance is strict because it is needed in the real world to keep voice quality acceptable.

The `DataTrafficGenerator.ned` module makes fake data traffic that looks like normal file transfers and web browsing. This traffic is for best-effort apps that don't need strict quality standards. The packets are medium-sized (1000 bytes) and the bitrate is moderate (1 Megabit per second). This traffic comes in bursts that last for 5 seconds, which is similar to how many data applications work: they turn on and off.

3.4 Performance Monitoring

The `PerformanceMonitor.ned` module has full network performance measurement features. This module collects statistics on throughput, latency, and packet loss across the entire network all the time, with a measurement interval of 1 second. The convergence detection mechanism keeps an eye on the stability of the routing table. It uses a configurable threshold of 0.1 to figure out when the network has reached a stable routing state after changes to the topology or initialization.

Performance data is saved in both vector files, which are used for time-series analysis, and scalar files, which are used for aggregate statistics. The output files `performance_results.vec` and `performance_stats.sca` have all the information you need to process and display the data. This data is what we use to compare the OSPF-TE and RIP routing protocols.

3.5 Implementation Files

The OSPF-TE Router implementation has header and source files that make the full OSPF-TE routing function work. The `TrafficEngineeringDatabase` class keeps track of all the network links and stores information about their bandwidth availability, propagation delays, current load levels, and calculated traffic engineering metrics. This database gets new information from the router's interfaces and link state advertisements from other routers on a regular basis.

The `ConstraintBasedRouting` class uses advanced algorithms to figure out the best path that take into account both traditional routing metrics and quality-of-service constraints. This module checks potential paths against minimum bandwidth and maximum delay requirements when figuring out how to get to a destination. It throws out paths that don't meet these requirements. The algorithm chooses the path with the best composite metric from a list of paths that meet the requirements. This metric is based on weighted combinations of bandwidth, delay, and load factors.

To get a single cost value for each link, the routing metric calculation uses a number of different factors. Bandwidth has an inverse effect on the metric, so links with more capacity cost less. Delay is directly related to the metric, and links with higher latency pay more for the same amount of data. The current link load gives the routing protocol real-time information about how much of the network is being used, so it can avoid busy paths when there are other options. You can change the relative importance of these factors by changing the weight parameters, which lets you optimize for different operational situations.

Link state advertisement handling uses the flooding method that is the most important part of OSPF. When a router's local link state changes a lot, it makes a new LSA with new information about its directly connected links. Before being sent to all nearby routers, this LSA gets a sequence number and a timestamp. The LSA is processed by the recipients, who then update their link state databases and send it on to their own neighbors. This makes sure that topology information is sent to everyone on the network. Duplicate detection systems stop LSA loops and cut down on the extra work that comes from processing duplicate data.

The RIP Router implementation uses a different distance-vector routing method. RIP routers keep simple routing tables that keep track of the destination network, the next hop router, the hop count metric, and the time stamp for each route. Periodic routing updates send full routing table information to neighbors that are directly connected. When routers get a RIP update, they compare the advertised routes to their current routing table and update the entries when they find shorter paths or when existing routes become invalid. This method is easier to use, which means it takes less processing power, but it doesn't make routing decisions as smartly as OSPF-TE does.

Traffic generator implementations make packet streams that look like the types of applications they are used with. The voice generator makes small packets with strict timing needs, the video generator makes constant bitrate traffic with large packets, and the data generator makes bursty on-off patterns. Each generator keeps track of its own internal state for packet sequencing and timing. This makes sure that the traffic it generates is a good representation of how the real application works.

The performance monitor implementation keeps track of network-wide statistics by intercepting or getting copies of packets at key points in the topology. Throughput calculations add up the bits sent over a certain amount of time, giving you time-series data on how much the network is being used. Latency measurements keep track of the time stamps on packets from when they are created to when they are received. This includes delays in propagation, transmission, and queuing. Packet loss detection looks at the number of packets sent and the number of packets received to find out where and when network congestion or routing problems cause packets to be dropped.

4 Configuration and Scenarios

The `omnetpp.ini` file controls all of the simulation settings and sets up several different test cases. General simulation settings set a 100-second time limit for the simulation and a 300-second time limit for the CPU to keep simulations from getting out of control. Random number generation uses configurable seeds to make sure that the same results can be gotten again and again, but that the results can change from one simulation run to the next.

The OSPF-TE router configuration gives each of the five routers a unique ID and lets them do Traffic Engineering. The metric weight parameters make sure that bandwidth, delay, and load factors all have an equal effect on path selection decisions. Default settings give bandwidth a weight of 1.0, delay a weight of 2.0, and load a weight of 1.5. This is because most people prefer low-latency paths but still want to take capacity and utilization into account. Quality of Service limits set minimum bandwidth requirements of 50 Megabits per second and maximum delay thresholds of 100 milliseconds. This makes sure that the chosen paths can handle applications that need high quality.

Protocol timing parameters control the frequency of hello messages, link state advertisements, and traffic engineering database updates. The set intervals find a balance between how quickly the protocol responds and how much extra data it needs to send. More frequent updates make it easier to adapt to changing conditions, but they also mean more control traffic. The hello interval of 10 seconds makes it possible to find out if a neighbor has failed fairly quickly, while the LSA interval of 30 seconds keeps the flooding of topology updates to levels that can be handled.

The standard RIP Version 2 settings for RIP router configuration include 30-second update intervals and 180-second route timeout periods. These conservative timer values show how RIP used to work in stable networks, where getting to the end of a route quickly was less important than keeping protocol overhead low. The maximum hop count of 15 is a basic limit of RIP, which means that the protocol can only work on networks with small diameters.

Setting up the traffic generator creates realistic traffic patterns for each application type of application. Video traffic works at 5 megabits per second, with 1500-byte packets sent every 30 frames per second. Voice traffic sends streams of 64 kilobits per second using 160-byte packets at 50 packets per second. Data traffic sends 1 Megabit per second in 1000-byte packets that burst every 5 seconds. These setups are like what real-world applications need and put different parts of network performance to the test.

A lot of different simulation scenarios make it possible to compare protocols in a systematic way across different operating conditions. The OSPF-TE Simulation scenario sets up a baseline for operation with balanced metric weights and moderate traffic loads. The RIP Comparison scenario sends the same traffic patterns through both routing domains so that performance can be measured directly. The High Load scenario doubles or triples the amount of traffic, which tests how the protocol works when there is a lot of traffic. The Low Latency scenario changes the weights of the metrics to put more emphasis on reducing delays, which is best for voice and video traffic. The High Throughput scenario focuses on making sure there is enough bandwidth, which speeds up data transfer. The Load Balancing scenario checks how well traffic is spread out over several available paths.

In parameter study configurations, some protocol parameters stay the same while others are changed in a systematic way. The Bandwidth Weight Study looks at how weighting bandwidth metrics affects path selection and overall performance. The TED Update Interval Study looks at the balance between how quickly routing responds and how much overhead the protocol adds. The Constraint Study looks at how different QoS constraint settings affect the availability of paths and the rate at which traffic is accepted. These studies give us a lot of information about how network behavior changes when protocol settings are changed.

5 Results and Analysis

The simulation results show that OSPF-TE and RIP routing protocols work very differently in a number of ways. A look at these results makes it clear what protocols should be used for different network needs and operational situations.

OSPF-TE showed better performance when there was a lot of traffic. When network traffic got close to or over the available capacity, the traffic engineering features made it possible to intelligently spread the load across several available paths. The constraint-based routing system was able to find paths that met bandwidth and delay needs while still providing good service for priority traffic, even as the overall network usage went up. This behavior is very different from RIP's hop-count-based routing, which often chose paths that weren't the best and got crowded while other paths stayed underused.

Latency tests showed another big benefit of using OSPF-TE. OSPF-TE routers always chose low-latency paths for traffic that needed to be delayed because they took propagation delay into account when making path selection decisions and enforced maximum delay limits. Voice and video streams had shorter and more stable end-to-end delays than similar traffic that went through the RIP network. OSPF-TE routing was able to adapt to changing network conditions because it was dynamic. It could reroute traffic away from busy paths that had longer queueing delays.

Measurements of convergence time showed that there were big differences between the protocols. The link-state architecture of OSPF-TE made it possible to quickly spread topology information across the entire network by flooding LSAs. After initializing the topology or making simulated changes to it, OSPF-TE routers got stable routing tables much faster than RIP routers. RIP's distance-vector structure meant that routing information had to go through several update cycles before it could be sent across the network. This led to long periods of convergence during which routing decisions were still wrong or not optimal.

Support for Quality of Service became a key benefit of OSPF-TE. When figuring out paths, the constraint-based routing mechanism took into account bandwidth and delay needs. This made sure that QoS-sensitive flows got the right network treatment. When routed through the OSPF-TE network instead of the RIP network, video streaming traffic always had a higher throughput and a lower packet loss rate. Voice traffic did better because there was less jitter and latency that was easier to predict. Best-effort data traffic worked well with both protocols, which is what you would expect for apps that don't have strict quality standards.

Parameter sensitivity analysis provided significant insights for the optimization of the protocol. The bandwidth weight parameter had a big effect on choosing paths for high-volume traffic flows. When the bandwidth weight went up, it made better use of high-capacity links. The delay weight parameter was very important for the quality of voice traffic. A higher delay weight ensured that low-latency paths were chosen, even if they had less capacity. The load weight parameter had an effect on how load balancing worked. Higher values made traffic spread more evenly across all available paths.

The frequency of updates to the Traffic Engineering Database showed that there are trade-offs between how quickly routing responds and how much overhead the protocol has. More frequent TED updates made it easier to adapt to changing network conditions, but they also caused more control traffic. The default update interval of 5 seconds worked well for the simulated scenarios, but the best settings may be different for different types of networks and traffic patterns. Updates that happened very often didn't help as much as they could have because routing decisions couldn't fully stabilize between updates.

The comparative analysis of various traffic types elucidated protocol behavior in response to varying application requirements. OSPF-TE's bandwidth-aware routing helped video traffic a lot because it needs a lot of bandwidth and can tolerate a little delay. The protocol was able to find and use high-capacity paths, which kept the streaming quality high even when there was a lot of traffic. Voice traffic with strict delay requirements showed how useful delay-weighted routing and maximum delay limits can be. OSPF-TE kept voice quality that would have gotten worse with RIP's hop-count-based routing by taking latency into account when choosing a path. The performance of data traffic showed less dramatic differences between protocols because best-effort applications can handle variable performance without major quality loss.

6 Discussion

The experimental results and observations facilitate a thorough assessment of both routing protocols and their appropriateness for various deployment contexts. Knowing what these protocols are like, what they can do, and what they can't do will help you make smart choices about network architecture and which routing protocol to use.

The fact that OSPF-TE includes Traffic Engineering sets it apart from other routing protocols. The Traffic Engineering Database keeps track of all the details about network link states, such as basic connectivity, available bandwidth, propagation delays, and current utilization levels. This detailed information lets you make routing decisions based on the current state of the network instead of just the static configuration. Routers can find links that are too busy and send traffic around them, which makes the whole network work better and applications run faster. Because TED updates are always changing, routing decisions are always based on the most up-to-date information. However, the frequency of updates must be carefully set so that responsiveness is balanced with protocol overhead.

Constraint-Based Routing is a big step forward from the old way of finding the shortest path. CBR makes sure that the paths it chooses can actually handle the traffic they carry by taking into account bandwidth and delay needs when calculating paths. This feature is very important for networks that support many different types of applications with different quality needs. For real-time apps like voice and video, paths need to meet strict latency and jitter limits. For bulk data transfers, high-capacity paths are better even if they add a little more time. Traditional routing protocols can't tell the difference between these two things, which means they might choose paths that connect the source and destination but don't meet the needs of the application.

The link-state architecture of OSPF-TE allows for faster convergence, which gives it operational benefits that go beyond just comparing protocols. In production networks, link failures, equipment maintenance, or capacity upgrades can change the topology. Routing protocols must quickly set up new stable routing states to handle these changes. Longer convergence periods create times when routing decisions might be wrong or not the best, which could lead to service interruptions or worse performance. OSPF-TE's fast convergence makes these windows smaller, which makes the network more reliable and lessens the effects of changes in topology.

RIP is simple, which can be helpful in some situations, even though it doesn't work very well. The protocol's low processing and memory needs make it a good choice for environments with limited resources, such as routers with limited processing power or memory. The straightforward distance-vector algorithm is easier to implement than link-state protocols, which have complicated SPF calculations and database management. It may be easier for small networks or organizations with limited networking knowledge to set up. But these benefits become less important as the network gets bigger and the needs of the applications get more demanding.

The limitations seen in RIP are due to basic features of distance-vector routing. Metrics that are based on hop count don't take into account link capacity or how much it is being used right now often chooses paths that go through more powerful links just because they require fewer router hops. The periodic full-table exchange creates update patterns that are predictable but not flexible. This means that it can't quickly respond to changes in the network or only update routes that have been changed. The limit on the maximum number of hops makes it hard to deploy in larger networks. These built-in limitations make RIP less and less useful for today's networks, which need to support a wide range of applications with strict quality standards.

Traffic type analysis showed how the features of different applications work with the capabilities of routing protocols. Intelligent path selection that takes into account available capacity is very important for high-bandwidth apps like video streaming. OSPF-TE's bandwidth-aware routing always chose high-capacity paths, which kept the quality of streaming high. RIP's capacity-blind routing, on the other hand, made streaming quality worse. VoIP and other low-latency applications showed how important it is to have routing that takes delays into account. By clearly limiting and reducing end-to-end delay, OSPF-TE kept voice quality that would have been hurt by other routing methods. Best-effort data applications were less affected by the choice of routing protocol because they could work well with different routing strategies because they could handle performance that changed.

The performance of routing protocols is greatly affected by the design of network topologies. The OSPF-TE mesh topology had many paths that connected to each other, which let the protocol use its advanced path selection features. Alternative routes gave people choices for load balancing and avoiding congestion, which turned advanced protocol features into real performance gains. The RIP network, on the other hand, had a simpler topology with fewer alternative paths, which made it harder to make smart routing decisions. This observation implies that the implementation of advanced routing protocols should be accompanied by topology designs that ensure adequate path diversity to leverage protocol functionalities.

The different configuration scenarios showed how the protocol worked when different operational needs were met. High-load situations put the protocols' ability to keep performance up under congestion to the test, showing that OSPF-TE is better at distributing load. Low-latency scenarios showed how important it is for real-time applications to use delay-aware routing. High-throughput situations proved that bandwidth-aware path selection is the best way to get the most data transfer rates. Load-balancing scenarios demonstrated the efficacy of load-weighted routing metrics in traffic distribution across multiple paths. Each scenario offered distinct insights into protocol optimization tailored to specific operational objectives.

7 Limitations and Future Directions

The current implementation has a few flaws that point to ways it could be improved and studied in the future. The routing protocol implementations are functional and representative, but they make some things easier than full IETF standard specifications. Instead of using the full Type 1 through Type 5 hierarchy, OSPF-TE LSA types have been simplified. RIP implementation focuses on the main functions and leaves out some of the optional ones in RIP Version 2. These simplifications make it easier to understand showing of basic ideas, but it might not cover all the subtleties of how to set up production protocols.

With only nine routers in the network, it's hard to draw conclusions about how well protocols work in big deployments. Real-world service provider networks may have hundreds or thousands of routers, which makes it hard to fully understand routing table sizes, convergence times, and protocol overhead in smaller topologies. Future research should expand the simulation to larger scales, examining the variations in protocol performance and resource requirements as network size increases.

Not having link failure scenarios in standard configurations is a big mistake. Link and node failures happen all the time in production networks because of broken equipment, software bugs, fiber cuts, and many other things. Routing protocols need to find failures, spread information about them, and quickly find new paths to minimize service interruptions. Future improvements should include failure scenarios that track how long it takes to find a failure, how quickly it converges, and how available the service is during failure and recovery periods.

The current system does not have any security features, such as ways to verify and give permission. Route injection, LSA flooding, and man-in-the-middle attacks are just a few of the attacks that production routing protocols must protect against. OSPF allows protocol messages to be authenticated with cryptography, and there are many ways to make RIP deployments safe. Future work ought to integrate these security mechanisms, assessing their influence on protocol performance and efficacy against particular attack scenarios.

The possibility of using MPLS (Multi-Protocol Label Switching) to improve Traffic Engineering is an exciting new direction for the future. MPLS allows for clear path specification and traffic engineering that goes beyond what pure IP routing can do. OSPF-TE for topology discovery and constraint-based path computation, along with MPLS for explicit path forwarding, could make traffic engineering even more advanced. This integration would show how modern protocol stacks bring together different technologies to make networking more powerful.

Adding more complex application behaviors to the traffic models would make the simulation more realistic. Current traffic generators make patterns that aren't very complicated, but real applications have more complicated behaviors, like changing bitrates, adaptive quality mechanisms, and complicated protocol interactions. Adding more realistic application models, like TCP congestion control, adaptive video streaming, and realistic web browsing patterns, would put the routing protocols to the test in more realistic situations.

8 Conclusion

This project successfully set up and tested OSPF, Traffic Engineering, and RIP routing protocols in a full network simulation environment. The experimental results show that OSPF-TE is clearly better for modern networks that need Quality of Service guarantees, traffic engineering capabilities, and high-performance operation. We have measured the performance improvements that can be made with advanced routing protocols and traffic engineering techniques by testing them in a variety of situations and with different types of traffic.

The implementation successfully achieved essential OSPF-TE functionalities, including Traffic Engineering Database management, Constraint-Based Routing for QoS-aware path selection, and link-state advertisement flooding for topology distribution. The implementation of RIP gave us a way to compare how distance-vector routing works in real life. The use of realistic traffic generators for video, voice, and data applications made it possible to test in conditions that were similar to those found in real-world network deployments. Comprehensive performance monitoring across all scenarios yielded the empirical data essential for thorough protocol comparison.

The results confirm Traffic Engineering principles, showing quantifiable enhancements in network utilization, application quality of service, and operational efficiency. Routing based on current network conditions always worked better than routing based on a fixed hop count. Constraint-based routing effectively upheld quality standards for delay-sensitive applications. The faster convergence of link-state routing made it easier to keep routing decisions stable, which was good for business. These results back up the idea that networks that support a wide range of applications with different quality needs should keep using advanced routing protocols.

Network designers should choose OSPF-TE for environments where application quality of service is important, where the network is too big for RIP, or where traffic engineering can help make the best use of the network. RIP is still useful for small, simple networks with low performance needs and limited resources. The results of the extensive parameter study can help you adjust protocol settings to get the best performance for certain operational scenarios, traffic mixes, and quality goals.

This simulation framework is great for teaching because it has a modular design and a lot of configuration options. Students can use it to learn about routing protocol operation, traffic engineering concepts, and network performance analysis. The implementation lays a strong groundwork for additional research into advanced networking subjects, including failure recovery mechanisms, security features, and integration with complementary technologies such as MPLS.

The knowledge gained from this work will help network protocols and architectures continue to change in the future. The proven advantages of traffic engineering and constraint-based routing underscore the significance of intelligent, adaptive network management. As networks continue growing in scale and supporting increasingly diverse applications with strict quality requirements, the need for sophisticated routing capabilities will only get worse. This project adds to the body of knowledge that supports this evolution by providing real-world proof of advanced routing protocol concepts and useful tips for how to use and improve them.

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